AZURE & MICROSOFT STACK

Autonomous HRO

The Agentic Hire-to-Retire Platform

An AI-native architecture where digital personas resolve HR work end-to-end — with humans in the loop.



Reference Architecture — One View

Azure & Microsoft stack · MCP + A2A native · numbered = request path

1 Experience — AI personas & proactive journeys — Teams · web · email · voice · joiner-mover-leaver event triggers — Copilot Studio · Foundry Agents

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2 AI Gateway authN/Z · rate limits · semantic cache · model routing · cost metering — APIM GenAI Gateway

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3 Multimodal Intake
docs, voice, video, images — Document Intelligence · Speech · GPT-4o

4 Intent, Priority & Routing
classify · SLA priority · route or escalate — Foundry classifiers · Service Bus

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5 Agent Orchestration Plane — A2A protocol · Semantic Kernel / Foundry Agent Service

- **Supervisor Agent** — plans, delegates to persona agents, enforces loop budgets (harness engineering)
- **Reason** — fast SLM path for repeat issues · CoT + mixture-of-experts for complex cases
- **Memory** — thread (short-term) · employee & case memory in Cosmos DB · vectors in AI Search
- **Human-in-the-Loop** — approval workbench · confidence thresholds · escalate to HR ops with full context

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6 MCP Services + Registry
tool & agent registry (discovery, versioning, RBAC) · policy RAG · curated HR knowledge base

6 Case Management & SLA
every request = auditable case · status, SLA timers, hand-offs between agents & humans

6 Act — Integration Layer
GET / PUT via APIM & Logic Apps connectors · approval-gated writes · idempotent retries

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7 Systems of Record — Workday · SAP SuccessFactors · payroll · benefits · ticketing · HR data stores

CROSS-CUTTING

Identity — humans & agents
Entra ID + Entra Agent ID · least-privilege, on-behalf-of · Key Vault

Guardrails
Content Safety · prompt shields · PII redaction · grounding checks

Observability & Evals
agent tracing (AgentOps) · continuous evaluation & red teaming · Azure Monitor

Governance & Cost
Purview data governance · model lifecycle · token cost / FinOps

Reporting
Microsoft Fabric · Power BI — deflection, SLA, quality, cost

Functional View — What Each Part Does

Same numbers as the technical view — ① here is ① there

1 Employee & Manager Front Door — one HR assistant wherever people work — chat, portal, email, phone — also reaches out first: onboarding, moves, exits

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2 Reception & Traffic Control — checks who is asking · keeps usage fair · tracks cost of every answer

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3 Understands Any Format
reads payslips, contracts, forms, ID scans · listens to voicemails · watches screen recordings

4 Triage Desk
works out what is being asked · how urgent it is · who (or what) should handle it

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5 Digital HR Case Team — works like a real HR team — a lead, specialists, and an escalation path

- **Case Lead** — owns the request end-to-end · splits work across specialist personas · stops runaway work
- **Judgment** — routine requests: instant playbook · complex cases: careful step-by-step reasoning
- **Remembers You** — knows the conversation so far · your history, country, grade, open cases
- **Asks a Human** — unsure or sensitive? hands the case to HR ops with the full story attached

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6 HR Policy Library & Toolbox
the HR policy database (per country, always current) · a catalog of approved actions agents may take

6 Case File & SLA Clock
every request becomes a tracked case — who did what, when, and is it still on time

6 Execution Desk
actually makes the change — updates records, triggers payroll — with approval where required

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7 Where HR Data Lives — HR record system · payroll · benefits · ticketing — unchanged, just connected

ALWAYS ON

Access Badges
people and agents only see what their role allows

Safety Net
blocks unsafe content · hides personal data · keeps answers on policy

Quality Control
every agent answer is logged, sampled and scored — like call QA

Oversight
data handling rules, audit trail, and cost per case under control

Dashboards
self-service rate, speed, satisfaction, cost — live for HR leadership

LAYER 1 · MEET THE EMPLOYEE

Agentic Experience & Multimodal Intake

Experience Layer

Named AI personas — "Mia" for payroll, "Leo" for onboarding — meet employees in Teams, web, email and voice for instant, autonomous resolution.

Azure AI Foundry Agent Service + Copilot Studio orchestrate the personas.

Multimodal Layer

Documents, voice files, videos and images flow in through Azure AI Document Intelligence, Speech-to-Text, Video Indexer and GPT-4o vision.

Every format becomes structured, actionable input.

Example An employee uploads a photographed marriage certificate plus a voice note — the platform reads both and triggers a life-event benefits change.

LAYER 2 · UNDERSTAND & CONVERSE

Intent Classification, Routing & Multi-Loop Conversations

Intent Layer

Classifies every request against the HRO intent catalog — payroll, leave, benefits, letters, exits — then prioritizes and routes via Azure AI Foundry classifiers and Service Bus.

Communication Layer

Multi-agent, multi-loop conversations over the A2A (agent-to-agent) protocol, with human-in-the-loop checkpoints for approvals and escalations.

Loop & Harness Engineering

Bounded agent loops, retry budgets and evaluation harnesses keep conversations convergent — never runaway.

Example "My bonus looks wrong" → payroll dispute, priority high → Payroll Agent loops with Compliance Agent, then asks the HR manager to approve the correction.

LAYER 3 · THE AGENT'S BRAIN AND HANDS

Services, Reasoning & Action

SERVICES · MCP

Tools, resources and prompts managed via Model Context Protocol servers — policy RAG on Azure AI Search, read-only access to HR knowledge and storage databases.

REASON

Fast-path reasoning for repeat issues; deep chain-of-thought and mixture-of-experts (o-series models) for novel, complex cases.

ACT

GET / PUT connectors into systems of record — Workday, SAP SuccessFactors, payroll engines, ticketing — via Azure API Management, Logic Apps and MCP tool calls.

Example "How many leave days do I have?" → instant RAG answer. "Relocation + tax + visa" → deep-reasoning agent drafts a full case and writes it back to the HCM.

LAYER 4 · TRUST FABRIC

Security, Guardrails & Observability

SECURITY

Microsoft Entra ID agent identities and RBAC, Key Vault, Purview data governance, private networking — every agent acts under least privilege.

GUARDRAILS

Azure AI Content Safety, prompt shields, PII redaction and policy-bound actions — writes always gated by human approval where required.

OBSERVABILITY

End-to-end tracing of every agent step in Azure Monitor and Application Insights; AgentOps evaluation and drift detection in AI Foundry.

Example An agent attempting to read salary data outside its scope is blocked, logged, and flagged to the security dashboard in real time.

LIFECYCLE FLOW 1 · HIGH VOLUME

Payslip Copy & Leave Balance

Repetitive

Deterministic — no reasoning at run time

AS-IS

Employee emails shared HR inbox → Ticket created & queued → Agent looks up payroll system → Replies with PDF → Ticket closed manually

1–2 days per request · thousands of identical requests every month

TO-BE

Employee asks the HR persona in Teams ① → Recognized as a known request type ④ → Approved tool fetches the payslip ⑥ → Straight from payroll ⑦ → Delivered in chat · case auto-logged ⑥

Seconds instead of days · zero human touch · every retrieval audited

①–⑦ = component on the architecture slides · amber = human in the loop

LIFECYCLE FLOW 2 · KNOWLEDGE

Policy Question: ”Can I carry over unused vacation?”

Grounded in HR policy database

Cited answers

AS-IS

Employee searches the intranet → Finds outdated pages → Emails the HRBP → Waits for a reply →

Answers differ by person & country

Days to answer · inconsistent guidance creates compliance risk

TO-BE

Ask in any channel ① → Triage identifies a policy question ④ → Country-aware lookup in the HR policy library ⑥ →

Fast reasoning composes a cited answer ⑤ → Low confidence? routed to HR ops for review ⑤

Consistent, cited answers in seconds — one source of truth for every country

①–⑦ = component on the architecture slides · amber = human in the loop

LIFECYCLE FLOW 3 · COMPLEX CASE

Payroll Discrepancy Dispute

Multi-agent | Human-in-the-loop | Gated write-back

AS-IS

Ticket bounced payroll ↔ benefits ↔ HR → Employee resubmits evidence → Manual recalculation → Correction lands next pay cycle

2–4 weeks · repeated hand-offs · frequent escalations to management

TO-BE

Employee uploads a payslip photo ①③ → Case lead delegates: payroll agent recomputes, benefits agent checks deductions ⑤ →

Agents reconcile findings agent-to-agent ⑤ → Proposed correction sent to a human approver ⑤ → Approved fix written to payroll ⑥⑦ →

Employee notified · SLA case closed ⑥

Hours instead of weeks — the human approves the fix, never re-keys the data

①–⑦ = component on the architecture slides · amber = human in the loop

LIFECYCLE FLOW 4 · FRONTIER

Cross-Border Relocation (Mover)

Beyond traditional RPA / chatbots

Deep reasoning

Multi-agent + human gates

AS-IS

HRBP + tax advisor + immigration firm trade emails → Spreadsheets track 40+ steps → Employee chases status for months

8–12 weeks of fragmented hand-offs — too much judgment for classic automation

TO-BE

Mover event proactively opens the case ① → Visa, contract & tax documents read on intake ③ →

Deep reasoning untangles tax, comp & benefits interactions ⑤ → Immigration, comp & payroll agents coordinate the plan ⑤ →

Human sign-off at each critical gate ⑤ → Changes executed across all systems ⑥⑦ → Employee sees one live status in chat ①

Weeks instead of months — humans keep the judgment calls, agents do the coordination

①–⑦ = component on the architecture slides · amber = human in the loop

From Solution to Product — Multi-Tenant by Design

one product · many clients · hard isolation where it matters

Product Control Plane — shared, run by the provider

one codebase, one release train, per-tenant configuration

Tenant Onboarding Factory
provision a new client in days: config, connectors, policy import

Release Rings
canary → early adopters → all tenants · feature flags per tenant

Shared Model Pool
central model catalog & upgrades · per-tenant routing and pinning

Central Evals & Guardrails
baseline safety and quality bars no tenant can lower

Metering & Billing
per-tenant usage, token cost, SLA credits, invoicing

Client A

Retail · 40k employees

- Branded personas & channels — client look, tone, languages
- Own policy library, intents & playbooks — per country, per union agreement
- Isolated data, memory & case files — encrypted per tenant, own keys
- Own HCM connectors — Workday here, SuccessFactors there

Client B

Manufacturing · 12 countries

- Branded personas & channels — client look, tone, languages
- Own policy library, intents & playbooks — per country, per union agreement
- Isolated data, memory & case files — encrypted per tenant, own keys
- Own HCM connectors — Workday here, SuccessFactors there

Client C

Financial services · regulated

- Branded personas & channels — client look, tone, languages
- Own policy library, intents & playbooks — per country, per union agreement
- Isolated data, memory & case files — encrypted per tenant, own keys
- Own HCM connectors — Workday here, SuccessFactors there

Shared Foundation

agent runtime · MCP registry · gateway · observability — tenant-tagged end-to-end, noisy-neighbor quotas per tenant

Isolation choices per tier: shared-with-row-level-isolation → dedicated database → fully dedicated deployment (regulated clients)

Running It as a Product — What Else Must Be True

the checklist that separates a client solution from a sellable platform

Isolation & Trust

Tenant isolation tiers — pooled, siloed DB, or dedicated deployment — priced accordingly

Per-tenant encryption keys — client-managed keys; agents can never cross tenant boundaries

No cross-tenant learning — one client’s data never tunes models or prompts for another

Compliance & Residency

Data residency — EU data stays in EU regions — deployable per geography

Certifications — SOC 2, ISO 27001, GDPR; AI-specific: EU AI Act readiness

Works councils & unions — per-country consent, transparency and audit obligations

Commercial Model

Pricing & metering — per employee served / per resolved case — with token cost visibility

SLAs per tier — response, resolution and uptime commitments with credits

Cost governance — per-tenant budgets and alerts so one client can’t burn the margin

Operations at Scale

Tenant admin console — clients manage their own policies, personas, approvers, intents

Per-tenant quality — evals, deflection and satisfaction tracked per client, not just globally

Incident model — tenant-aware on-call, status pages and breach notification duties

Customization vs Core

Config, not forks — 80% product, 20% per-client config: intents, playbooks, connectors

Connector marketplace — certified HCM / payroll / ticketing connectors, versioned per tenant

Upgrade safety — tenant-level regression evals before any model or prompt rollout

Go-to-Market

Onboarding factory — policy import, connector setup, HITL role mapping — days not months

Trust artifacts — model cards, audit logs, explainability reports for client security reviews

Exit & portability — clients can leave with their data, cases and policy library intact

Choosing an Isolation Tier — Cost vs Control

one product, three deployment shapes — sold as tiers

	Tier 1 · Pooled Shared everything, row-level isolation	Tier 2 · Siloed Data Shared app, dedicated database & storage	Tier 3 · Dedicated Fully dedicated deployment
WHAT IS SHARED	Shared app, database & model pool — every row tenant-tagged, enforced at the query layer	Shared runtime & release train — but each client gets their own database, vector store & case files	Entire stack — runtime, models, gateway, data — deployed into a dedicated (or client-owned) cloud subscription
SECURITY & KEYS	Encryption per tenant · strict RBAC · noisy-neighbor quotas	Client-managed encryption keys (BYOK) · isolated backups & retention	Client controls keys, network, private endpoints · optional air-gap from provider
COMPLIANCE FIT	Standard compliance (SOC 2, GDPR) — no special residency demands	Meets data residency — database pinned to client's region	Regulated industries: banking, insurance, public sector · auditor access
COST PROFILE	Lowest cost per employee · best margin	Moderate cost premium · priced into the tier	Highest cost · premium pricing & minimum commitments
SPEED & UPGRADES	Onboarding in days · instant upgrades for all	Onboarding in 1–2 weeks · upgrades per release ring	Onboarding in weeks · upgrades scheduled with the client
TYPICAL CLIENT	Best for: mid-market clients with standard requirements	Best for: enterprises with residency or works-council demands	Best for: regulated clients where shared infrastructure is a non-starter

Same codebase and release train in every tier — only the deployment boundary changes

SaaS or On-Prem? — Operated as SaaS in Every Tier

the real question is where the boundary of your data sits

Why the platform must be operated as SaaS

The product improves weekly — agent quality depends on continuous evals, prompt & guardrail updates — a frozen on-prem install degrades from day one

Frontier models don't live on-prem — deep reasoning needs top-tier Azure-served models; disconnected installs are limited to weaker local models

Safety is a 24/7 operation — red teaming, jailbreak patching and incident response are provider duties — not a client IT workload

Economics only work pooled — shared model pool, gateway and runtime are what make per-employee pricing viable

What "we need on-prem" usually means — and the answer

"Our data can't leave" — Tier 2/3: data pinned to your region or your own Azure subscription — provider never holds it pooled

"We control the keys" — BYOK / client-managed keys in every dedicated tier · private endpoints, no public network path

"Auditors must inspect it" — dedicated tier grants auditor access, full audit trail, model cards & explainability reports

"No shared infrastructure" — Tier 3 deploys the entire stack into a client-owned subscription — private SaaS, still provider-operated

The legitimate hybrid — split data plane

defuses data sovereignty without forking the product

Client environment
HR data, case files, memory, policy library, HCM connectors — everything stateful stays inside the client boundary

PII redacted at the boundary →
← answers only

Provider model pool
stateless reasoning calls only — no retention, no training on client data, frontier models always current

Position: SaaS-first · "private SaaS in your cloud" for regulated clients · classic disconnected on-prem is a worse product, not a cheaper one

Hardware, Software & Infrastructure Requirements

everything runs on Azure — clients bring identity, channels & system access

Hardware & Capacity

No client hardware in SaaS tiers
all compute is cloud-hosted; employees only need a browser or Teams

Model capacity
Azure OpenAI provisioned throughput (PTU) for predictable latency · pay-as-you-go burst for peaks

App compute
AKS / Container Apps — autoscaling agent runtime, ~2–4 vCPU per 1k active employees as a starting point

GPU
none owned — GPU is consumed via the model services, never managed directly

Dedicated tier sizing
sized per client: employee count, case volume, document OCR load

Software Stack

AI platform
Azure AI Foundry (agents, evals) · Azure OpenAI · Document Intelligence · Speech · Content Safety

Runtime & orchestration
Semantic Kernel / Foundry Agent Service · MCP servers · A2A messaging on Service Bus

Data layer
Cosmos DB (memory & cases) · AI Search (vectors & policy index) · Blob storage (documents)

Integration
API Management (gateway) · Logic Apps connectors · certified HCM connector pack

Insight & ops
Microsoft Fabric + Power BI · Azure Monitor + App Insights · IaC via Bicep/Terraform

Infra & Network

Landing zone
per-region Azure landing zone · tenant-tagged resources · release rings across regions

Network
private endpoints & VNet integration · no public data path in Tier 2/3 · TLS everywhere

Identity
Entra ID federation with client IdP (SSO) · Entra Agent ID for every agent · Key Vault / BYOK

Resilience
zone-redundant services · cross-region DR for dedicated tiers · RPO/RTO per SLA tier

Environments
dev / staging / prod per release ring · tenant sandbox for client UAT before go-live

Client-Side Prerequisites

Channels
Microsoft 365 with Teams (or web portal fallback) · mail routing for the email channel

Identity
Entra ID (or SAML/OIDC IdP) for SSO · agreed role mapping for HITL approvers

System access
API credentials for HCM, payroll, benefits, ticketing — least-privilege service accounts

Data feeds
HR policy documents per country · org & employee master data sync

Network
egress allow-listing or private connectivity (ExpressRoute / VPN) for Tier 2/3

Pricing Models & Options

price the outcome, meter the cost, protect the margin

PEPM Subscription

Primary model

Per employee served, per month — all-you-can-ask within fair use

· Predictable for CHRO budgets · scales with workforce size

· Includes standard flows, policy Q&A, case handling

· Fair-use token cap with transparent overage

Best for: enterprise-wide deployments

Per Resolved Case

Outcome-based

Pay only when the platform fully resolves a case end-to-end

· Zero-risk entry — aligned to deflection value

· Human-escalated cases billed at a reduced rate

· Rate card by case complexity (Flow 1 ≠ Flow 4)

Best for: pilots & value-proof phases

Platform + Usage

Dedicated tiers

Annual platform fee + metered usage (tokens, cases, storage)

· Covers dedicated infrastructure & compliance overhead

· Minimum commitment · multi-year discounts

· Client sees the same cost telemetry we do

Best for: Tier 3 regulated clients

Tier multipliers & add-ons

Isolation tier uplift — Pooled 1.0× · Siloed Data ~1.3–1.5× · Dedicated ~2–3× (infra passed through + ops premium)

Add-ons — extra HCM connectors · premium deep-reasoning quota · additional languages · managed HITL desk · advanced analytics pack

Onboarding — one-time implementation fee, credited back on multi-year commitment

Commercial guardrails

Token economics — fair-use bands per employee; model cost pass-through above band keeps margin safe as usage grows

SLA credits — uptime & resolution-time commitments per tier, credited automatically from metering data

Value ratchet — annual price review indexed to measured deflection & satisfaction, not seat count alone

Implementation Plan — ~26 Weeks to Full Rollout

per client onboarding · phases overlap deliberately



Team: solution architect · agent engineers ×3 · integration engineers ×2 · HR process lead · security engineer · client-side SMEs — subsequent clients compress to ~10–14 weeks via the onboarding factory

Legal & Product Documentation — Post-Release

the right document, at the right lifecycle stage

Contract & Onboard

signature → go-live

MSA / Terms of Service
the master contract — liability, IP, warranties, termination

Order Form / SOW
scope, tiers, pricing, onboarding deliverables

Acceptable Use Policy
what customers can & can't do — no scraping, misuse, unlawful content

DPA + SCCs
GDPR data processing terms, cross-border transfer clauses

SLA Schedule
uptime & resolution commitments, credit mechanics

Live Operations

continuous, in-product

Privacy Notice
what employee data is processed, why, and for how long

AI Transparency Notice
EU AI Act — users know they interact with AI; HITL disclosed

AI Output Disclaimer
answers are policy-grounded assistance, not legal/tax advice

Subprocessor List
Azure, model providers — with change-notification rights

Trust Center Pack
SOC 2 / ISO 27001 reports, pen-test summaries under NDA

Change & Update

per release / model change

Release Notes
features, fixes, deprecations — functional documentation

Change Notifications
material changes to terms, models, or subprocessors

Model / Feature Cards
what changed in AI behaviour, evals run, known limits

Updated User Guides
admin & end-user documentation kept in step

Incident & Exit

event-driven / renewal

Incident Notification
breach notice within contractual/legal windows (72h GDPR)

RCA / Post-Incident Report
root cause, impact, remediation for affected tenants

Audit & Compliance Reports
annual attestations, regulator-facing evidence

Exit & Data Return Plan
export formats, deletion certificate, transition assistance

Operating principle — contractual documents (MSA, DPA, SLA) are versioned and countersigned; in-product notices (AUP, privacy, AI transparency) are always-current and linked from the assistant itself; event-driven documents (incidents, RCAs, exit plans) have contractual clocks. One legal register tracks which version each tenant is on.

LAYER 5 · INSIGHT & IMPACT

Reporting, Dashboards & the Road Ahead

70–80%

of routine HRO intents resolved autonomously — minutes, not days (illustrative targets)

Microsoft Fabric + Power BI dashboards track resolution rates, autonomy share, CSAT, cost per case and human-in-loop interventions.

MCP and A2A open standards mean new agents, tools and vendors plug in without re-architecture.

Start with 2–3 high-volume intents. Measure. Then scale the agent mesh across the full hire-to-retire cycle.